

**B.TECH
(SEM III) THEORY EXAMINATION 2022-23
FLUID MECHANICS**

Time: 3 Hours**Total Marks: 70****Note:** Attempt all Sections. If require any missing data; then choose suitably.**SECTION A****1. Attempt all questions in brief.****2 x 7 = 14**

- (a) Explain buoyancy and center of pressure.
- (b) Define velocity potential function.
- (c) What is the difference between notches and weir?
- (d) List various types of minor losses.
- (e) Explain boundary layer thickness with diagram.
- (f) What are distorted models?
- (g) Define dynamic similarity.

SECTION B**2. Attempt any three of the following:****7 x 3 = 21**

- (a) Derive an equation to calculate total force acting on inclined plane.
- (b) Define stream function & velocity potential function. Calculate for the velocity at the point (1,2) for the following stream function: $\psi = 0.6 (y^2 - 3x^2) + xy - 6$.
- (c) What is orificemeter? Derive an equation to calculate the discharge using orificemeter.
- (d) What is laminar sub-layer. Calculate displacement thickness for velocity distribution in boundary layer given by: $\frac{u}{U} = (\frac{y}{\delta})^2$?
- (e) Define various dimensionless number and write its significance.

SECTION C**3. Attempt any one part of the following:****7 x 1 = 7**

- (a) Draw the Rheological graph of various fluids. Also explain rheological behavior of fluids.
- (b) What are manometers? Explain vertical single column manometer.

4. Attempt any one part of the following:**7 x 1 = 7**

- (a) Derive the general form of continuity equation for 3-D flow. Also write its assumptions.
- (b) Following data are given:- Barometer reading = 740 mm of mercury, specific gravity of mercury = 13.6, intensity of pressure = 40 kPa. Calculate the intensity of pressure in S.I. units, both gauge and absolute.

5. Attempt any *one* part of the following:

7 x 1 = 7

- A 30 cm diameter horizontal pipe terminates in a nozzle with the exit diameter
- (a) of 7.51 cm if the water flows through the pipe at a rate of 0.15 m³/sec. Calculate the force that will be exerted by the fluid on the nozzle?
 - (b) Explain loss of energy in pipe. Enlist various types of minor losses.

6. Attempt any *one* part of the following:

7 x 1 = 7

- The velocity distribution in boundary layer is given by: $(U/V) = (3\epsilon/2 - \epsilon^2/2)$. Calculate the boundary layer thickness, momentum thickness & energy thickness. Where, $\epsilon = \frac{\nu}{\delta}$
- (a)
 - (b) What is boundary layer separation. Draw a neat diagram to explain.

7. Attempt any *one* part of the following:

7 x 1 = 7

- The resulting force F of a supersonic plane during flight can be considered as dependent upon the length of aircraft "L", velocity "V", air viscosity " μ ", air density " ρ " and the bulk modulus of air "K". Express the functional relationship between these variables and the resisting force.
- A pipe of diameter 1.5 m is required to transport an oil of specific gravity 0.9, viscosity 3×10^{-2} poise, at the rate of 3000 liters/s. Test were conducted on a 15 cm diameter pipe using water at 20° C. Find the velocity and rate of flow in model. Viscosity of water at 20° C = 0.01 poise.
- (a)
 - (b)